ACTIVE ANODE

Information Bulletin

Electronically controlled cathodic corrosion protection for water heaters and industrial tanks

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Corrosion in water heaters

The permanent flow of fresh, oxygen-rich water will inevitably cause corrosion in a storage tank if no protective measures are taken.

Two traditional protection methods are used:

- **Enamel coating** — protects most of the surface, but micro-defects always remain unprotected.
- **Sacrificial magnesium anode** — dissolves instead of the tank, but once exhausted, protection ceases. Requires replacement every 1–3 years.

Solution: the active anode

The active (impressed current) anode generates a **constant electronically regulated protective current** via a titanium electrode. It does not deplete and requires no replacement.

Advantages of the active anode

✓ Does not deplete — lasts the full lifetime of the tank

✓ Automatically regulates the protective current

✓ Protects enamelled and stainless steel tanks

✓ Reduces scale on the heating element

✓ Installed in place of the standard magnesium anode

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How it works

The system consists of a **potentiostat** (electronic control unit) and a **titanium electrode** with mixed metal oxide (MMO) coating. Current is measured and applied at millisecond intervals:

1

Measurement

The potentiostat records the actual electrochemical potential inside the tank

2

Calculation

The required protective current to reach nominal potential is determined

3

Application

Protective current is fed via the titanium electrode. Tank and heating element become cathodes

4

Control

The cycle repeats continuously. Both over- and under-protection are automatically prevented

System components

Potentiostat

Electronic control unit. Connected to 230 V mains. Measures tank potential and regulates protective current in real time.

- Supply voltage: 230 V / 50–60 Hz
- Current: 50–180 mA
- Setpoint: 2.3 V (enamel) / 1.9 V (stainless)
- Ambient temperature: 0–40 °C
- Protection class: IP II

Titanium electrode (TA)

Made of titanium with MMO coating. Practically wear-free throughout the entire service life of the water heater.

- Material: Ti + MMO coating
- Lengths: 200 / 400 / 800 / 1200 mm
- Tank volume range: 50–5,000 L
- Service life: entire lifetime of the water heater
- Connection: SmartConnect plug (reverse polarity protected)

Comparison of corrosion protection methods

Criterion	Magnesium anode	Enamel	Active anode
Service life	Depletes (1–3 yr)	Full tank life	Full tank life
Maintenance	Regular replacement	None required	None required
Scale protection	Partial	No	Yes
Auto-regulation	No	N/A	Yes
Micro-defect cover	Partial	No	Yes
Tank type	Enamel / Stainless	Enamel only	Enamel + Stainless

System selection guide

The active anode is selected according to tank material, volume and number of heat exchangers.

Tank material	Volume	Setpoint	Electrode
Enamelled steel	50–800 L	2.3 V	1 × TA 200 or TA 400
Enamelled steel	800–2000 L	2.3 V	1–2 × TA 800
Stainless steel	50–500 L	1.9 V	1 × TA 400 or TA 800
Stainless steel	500–2000 L	1.9 V	2 × TA 800
Any, >2000 L	>2000 L	Custom	Selected after measurements

* TA = Titanium Anode. Exact selection is made after laboratory measurements of the specific tank. Height-to-diameter ratio must not exceed 3:1.

Installation and commissioning

Installation steps

1. Disconnect the water heater from the mains and drain some water.
2. Remove the existing magnesium anode from the threaded socket or flange.
3. Insert the titanium electrode (with adaptor if required).
4. Connect the electrode cable to the potentiostat.
5. Connect the potentiostat to the 230 V mains.
6. Check the operating indicator on the potentiostat display.

Mounting options

- **Socket mounting** (G¾″ or G1″) — most common, suits most water heaters.
- **Flange mounting** — for large industrial tanks.
- **Insulated-hole mounting** — for tanks without a dedicated anode port.

In tanks with multiple metallic components (heating elements, heat exchangers) all must be electrically insulated or fitted with balancing resistors.

Frequently asked questions

Does the active anode need to be replaced?	No. The titanium electrode is practically wear-free and lasts the entire lifetime of the water heater. Unlike a magnesium anode, it does not deplete.
Is the active anode compatible with any water heater?	Yes, provided a threaded socket or flange is available. The system suits tanks of enamelled or stainless steel with volumes from 50 to 5,000 litres.
Does it protect against scale?	Yes. The electrochemical reaction prevents the formation of hard $CaCO_3$ scale — a loose salt forms instead that does not adhere to the heating element.
What happens if the system is switched off?	Protection ceases and corrosion begins. Continuous operation is recommended.
Does the potentiostat require maintenance?	No. Only check the indicator LED once a year during routine inspection of the water heater.

All technical data is provided for general information purposes. Exact system component selection requires laboratory measurements of the specific tank.